

ANATOMY OF THE DEFORMED OSCILLATION FOR $h_n = \tau(p_n)$, AND THE QUESTION OF ZEROS

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ABSTRACT. This note records what can and cannot be said about the oscillation seen in the deformed minimum-modulus plots of the sequence $h_0 = 1$, $h_n = \lambda_n = \tau(p_n)$ ($n \geq 1$). Three things are established rigorously; one is supported by reliable computation at small n ; and the central law governing the long-range envelope is left open. The discussion separates these tiers carefully, and ends with the question of zeros, where there is a clean exact statement and a direct link to Lehmer's conjecture. Code is included.

1. WHAT IS ESTABLISHED

Notation. Let $h = (h_n)_{n \geq 0}$ with $h_0 = 1$ and, here, $h_n = \lambda_n := \tau(p_n)$, where τ is the Ramanujan tau function and p_n the n -th prime. Write $H(t) = \sum_{n \geq 0} h_n t^n$ and let the data sequence $j = (j_r)_{r \geq 1}$ be defined by $J(t) := \sum_{r \geq 1} j_r t^r = t \frac{d}{dt} \log H(t)$ (relation (D) of the manuscript). Theorem 2.3 of the manuscript gives the characteristic polynomial

$$\chi_n(x) = \sum_{r=0}^n \binom{n}{r} r! h_r (-1)^r x^{n-r},$$

equivalently,

$$\sum_{n \geq 0} \chi_n(x) \frac{t^n}{n!} = H(-t) e^{xt}.$$

The minimum modulus is $\mu_n := \min\{|x| : \chi_n(x) = 0\}$ (the plotted quantity); roots of χ_n are the eigenvalues of the manuscript's matrix. To *deform by* c is to prepend c to j , giving $\chi_n^{(c)}$ and $\mu_n^{(c)}$.

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(E1) Deformation is a change of generating function, then a shift.

Prepending c replaces $\log H = \int_0^t J/s \, ds$ by $\log H^{(c)} = ct + \int_0^t J \, ds$, so $H^{(c)}(t) = e^{ct}G(t)$ with

$$G(t) := \exp \int_0^t J(s) \, ds$$

independent of c . Letting Φ_n be collected by $\sum_n \Phi_n(x)t^n/n! = G(-t)e^{xt}$, the substitution $t \mapsto -t$ turns e^{ct} into e^{-ct} and yields the polynomial identity

$$\chi_n^{(c)}(x) = \Phi_n(x - c), \quad \text{hence} \quad \text{spec } \chi_n^{(c)} = c + \text{spec } \Phi_n.$$

The undeformed object uses H ; the deformed family uses the genuinely different backbone G (here $G'/G = J = t H'/H$). The oscillation lives in G , and c only translates the result.

(E2) The deformed minimum modulus is a distance. From (E1),

$$\mu_n^{(c)} = \text{dist}(-c, \text{spec } \Phi_n).$$

Computationally (Figure 1) the cloud $\text{spec } \Phi_n$ is a ring around an empty central disc, so $\mu_n^{(c)}$ measures the distance from the fixed point $-c$ to the ring's inner edge. The oscillation in n is the inner edge advancing toward and receding from $-c$.

(E3) An exact criterion for a zero. The constant term of χ_n is the $r = n$ term, $\chi_n(0) = (-1)^n n! h_n$, and since χ_n is monic the product of its roots is $\prod_i \rho_i = n! h_n$. Therefore, exactly,

$$\mu_n = 0 \iff \chi_n(0) = 0 \iff \tau(p_n) = 0,$$

and for the deformed family, with $h_n^{(c)} = [t^n] e^{ct}G(t) = \sum_{k=0}^n \frac{c^k}{k!} g_{n-k}$ (where $g_m = [t^m]G$),

$$\mu_n^{(c)} = 0 \iff h_n^{(c)} = 0.$$

These two identities were verified symbolically. They are the backbone of §4.

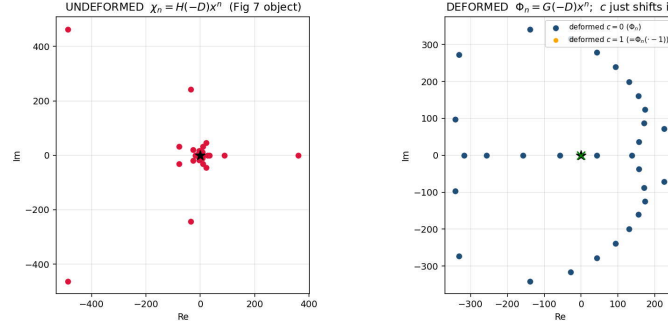


FIGURE 1. Root clouds for $h_n = \tau(p_n)$ at $n = 30$ (from a companion note). Left: the undeformed spectrum crowds the origin. Right: the deformed backbone G opens an empty disc ($\min |x| \approx 42$) and the eigenvalues lie on a ring; c merely shifts the ring.

2. THE ANATOMY OF THE OSCILLATION (COMPUTATIONAL, SMALL n)

Computed for $c = 1$ over consecutive $n = 2, \dots, 44$ (Figure 2; all values stable under doubling the working precision), the oscillation has a sharp internal structure.

- (1) *The nearest eigenvalue is real.* For $n \gtrsim 8$ the root of $\chi_n^{(1)}$ of least modulus lies on the real axis (argument 0 or π to machine accuracy), even though the spectrum is otherwise a complex ring. Equivalently, the inner edge of the ring touches nearest to $-c$ on the real axis.
- (2) *Period-four threads.* The sign of that nearest real root follows the pattern $(-, -, +, +)$ with period 4; equivalently $\mu_n^{(1)}$ splits into four smooth interleaved subsequences indexed by $n \bmod 4$ (the four colours in Figure 2). Each thread varies slowly and nearly monotonically across the computed range:

$$\begin{array}{ll}
 n \equiv 2 : & 25.6 \rightarrow 50.0 \ (\uparrow) \\
 n \equiv 3 : & 3.1 \rightarrow 27.0 \ (\uparrow) \\
 n \equiv 1 : & 40.3 \rightarrow 25.2 \ (\downarrow) \\
 n \equiv 0 : & 17.3 \rightarrow 0.4 \rightarrow 2.1 \ (\downarrow, \text{ then up}).
 \end{array}$$

- (3) *Fast oscillation vs. envelope.* The rapid up-down filling the band in the manuscript's Figure 2 is the interleaving of four threads sitting at very different heights; the slow scalloped envelope is the gradual drift of those threads. The dramatic dips toward 0 are one thread (here $n \equiv 0$) bottoming out.
- (4) *A genuine near-zero.* At $n = 40$, $\mu_{40}^{(1)} = 0.435$ — a real eigenvalue sitting very close to the origin. This value is unchanged when the precision is doubled, so it is a true feature: the deformed coefficient $h_{40}^{(1)}$ is small, not zero.

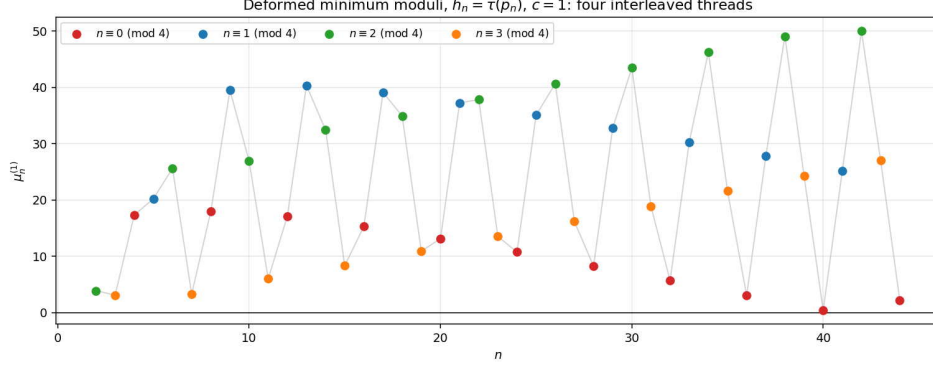


FIGURE 2. Deformed minimum moduli $\mu_n^{(1)}$ for $h_n = \tau(p_n)$, consecutive n , coloured by $n \bmod 4$. The fast oscillation is the interleaving of four slowly-drifting threads; the $n \equiv 0$ thread dives to the genuine near-zero $\mu_{40}^{(1)} = 0.435$.

I emphasise the status of §2: it is reliable computation over a short range ($n \leq 44$, the limit at which the characteristic polynomials are still well conditioned at feasible precision), not a theorem. The period-four real-root structure is robust in that range; whether it persists, and what governs the long envelope visible out to $n = 500$, I do not claim to know.

3. WHY THESE FEATURES, AND WHAT IS OPEN

The mechanism behind (E1)–(E2) is exact; the anatomy of §2 is not yet explained. The natural locus is the analytic structure of $G(t) = \exp \int_0^t J$. For the toy sequence $h \equiv 2$ one has $G = 1/(1 - t^2)$, whose two poles at $t = \pm 1$ (arguments $0, \pi$) produce a limit family $\cosh / \sinh$ with period-two (parity) behaviour. By analogy, a period-four pattern points to dominant singularities of G near the imaginary axis (arguments $\pm\pi/2$) in addition to the real ones, so that the inner edge of the ring crosses the real axis with a four-step rhythm. This is a hypothesis, not a result: for $h_n = \tau(p_n)$ the series H (hence G) has no closed form, $\tau(p_n)$ grows only polynomially in n (Deligne’s bound $|\tau(p)| \leq 2p^{11/2}$ gives radius of convergence 1), and the relevant singularities are not known to me. The honest reduction is: *the oscillation period equals the angular rhythm with which the inner edge of $\text{spec } \Phi_n$ meets the real axis, and that rhythm is fixed by the boundary singularities of G .*

4. THE QUESTION OF ZEROS

There are three distinct “zero” questions; (E3) keeps them apart.

Does the undeformed sequence vanish? — Lehmer’s conjecture.

By (E3), $\mu_n = 0$ if and only if $\tau(p_n) = 0$. Thus the undeformed minimum-modulus plot (manuscript Figure 7) touches 0 at n *exactly* when Ramanujan’s τ vanishes at the n -th prime. A single prime p with $\tau(p) = 0$ would force $\tau(pm) = 0$ for every m coprime to p , so non-vanishing over primes is the essential case of *Lehmer’s conjecture* that $\tau(n) \neq 0$ for all $n \geq 1$ [1]. That conjecture is open [4]; it has been verified computationally for all n far beyond anything in these plots (Lehmer’s own range and successors reach past $n = 2 \times 10^{11}$), and strong structural constraints are known — Deligne’s bound [2], Serre’s and Swinnerton-Dyer’s ℓ -adic theory and the congruences modulo 2, 3, 5, 7, 691 [3], and the recent result of Balakrishnan, Craig and Ono that $\tau(n) \notin \{\pm 1, \pm 3, \pm 5, \pm 7, \pm 691\}$ for $n > 1$ [4]. So: the undeformed ragged dips toward 0 in Figure 7 are, term by term, the assertion $\tau(p_n) \neq 0$; the plot cannot reach 0 unless Lehmer’s conjecture fails.

Does the deformed sequence vanish? By (E3), $\mu_n^{(c)} = 0$ if and only if $h_n^{(c)} = \sum_{k=0}^n \frac{c^k}{k!} g_{n-k} = 0$. This is a *different* arithmetic condition — a cancellation among the coefficients of $e^{ct}G$ — and is not controlled by Lehmer’s conjecture. For rational c (e.g. $c = 1$) the quantity $h_n^{(c)}$ is rational, so an exact zero would be an arithmetic coincidence; none was observed. What *is* observed (item 4 above) is genuine near-vanishing: $\mu_{40}^{(1)} = 0.435$. So the deformed plots can dip arbitrarily close to 0 for honest reasons, and the manuscript’s goal of bounding these eigenvalues away from 0 uniformly in n is the open quantitative form of the question — equivalently, a uniform lower bound on $|h_n^{(c)}|$, or on $\text{dist}(-c, \text{spec } \Phi_n)$.

Can a deformation remove the zeros of the undeformed problem?

Yes, in the precise sense of (E1): the undeformed spectrum reaches the origin only via the Lehmer mechanism, whereas the deformed spectrum sits on a ring of radius growing with n , so $-c$ is generically interior to the empty disc and $\mu_n^{(c)} > 0$ for structural (non-arithmetic) reasons. Deformation thus trades the Lehmer-type vanishing for the coefficient-cancellation vanishing of $h_n^{(c)}$ — which is why the deformed plots are bounded oscillations rather than ragged descents to 0.

5. CODE

The following is self-contained: it computes τ from $\Delta = q \prod (1 - q^n)^{24}$, forms the data sequence j , prepends c , builds $\chi_n^{(c)}$ from Theorem 2.3, and returns the minimum modulus together with the nearest root. It also checks the identity $\chi_n(0) = (-1)^n n! h_n$ of (E3).

```
import mpmath as mp
from math import comb, factorial
from fractions import Fraction as Fr
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```

def tau_table(M):
    # tau(1..M) from Delta = q*
    prod(1-q^n)^24
    prod = [0]*(M+1); prod[0] = 1
    for n in range(1, M+1):
        poly = [0]*(M+1); k = 0
        while n*k <= M and k <= 24:      # (1-q^n)^24 = sum C(24,k)
            (-1)^k q^{nk}
            poly[n*k] += comb(24, k)*((-1)**k); k += 1
        new = [0]*(M+1)
        for i in range(M+1):
            if prod[i]:
                for j in range(0, M+1-i):
                    if poly[j]: new[i+j] += prod[i]*poly[j]
        prod = new
    return lambda m: prod[m-1]          # tau(m) = [q^m] q*prod =
    prod[m-1]

def primes_upto(x):
    s = [True]*(x+1); s[0] = s[1] = False
    for i in range(2, int(x**0.5)+1):
        if s[i]:
            for j in range(i*i, x+1, i): s[j] = False
    return [i for i in range(2, x+1) if s[i]]

def j_from_h(h):
    # relation (D): n h_n =
    sum_{r=1}^n j_r h_{n-r}
    j = []
    for n in range(1, len(h)):
        s = sum(j[r-1]*h[n-r] for r in range(1, n))
        j.append(n*h[n] - s)
    return j

def h_from_j(j):
    # invert (D)
    H = [Fr(1)]
    for n in range(1, len(j)+1):
        s = sum(j[r-1]*H[n-r] for r in range(1, n+1))
        H.append(s/n)
    return H

def chi_coeffs(h, n):
    # Theorem 2.3, monic,
    descending powers
    return [Fr(comb(n, r))*Fr(factorial(r))*h[r]*((-1)**r) for r in
    range(n+1)]

def min_modulus(h, n, dps):
    mp.mp.dps = dps
    co = [mp.mpf(c.numerator)/mp.mpf(c.denominator) for c in
    chi_coeffs(h, n)]

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    roots = mp.polyroots(co, maxsteps=1500, extraprec=4*dps)
    z = min(roots, key=abs)
    return abs(z), z

M = 230
tau = tau_table(M)
P = primes_upto(M)
h = [Fr(1)] + [Fr(tau(p)) for p in P]    # h_0=1, h_n = tau(p_n)
j = j_from_h(h)
c = Fr(1)
hc = h_from_j([c] + j)                  # deform by c: prepend c
    to j

# exact zero criterion (E3): chi_n(0) == (-1)^n n! h_n
n = 20
assert chi_coeffs(hc, n)[-1] == Fr((-1)**n)*Fr(factorial(n))*hc[n]

for n in range(2, 45):
    mu, z = min_modulus(hc, n, 55 + 2*n)
    print(n, float(mu), complex(z))

```

REFERENCES

- [1] D. H. Lehmer, *The vanishing of Ramanujan's function $\tau(n)$* , Duke Math. J. **14** (1947), 429–433.
- [2] P. Deligne, *La conjecture de Weil. I*, Publ. Math. IHÉS **43** (1974), 273–307. (Gives $|\tau(p)| \leq 2p^{11/2}$.)
- [3] H. P. F. Swinnerton-Dyer, *On ℓ -adic representations and congruences for coefficients of modular forms*, in *Modular Functions of One Variable III*, Lecture Notes in Math. **350**, Springer, 1973, 1–55; see also J.-P. Serre, *Une interprétation des congruences relatives à la fonction τ de Ramanujan*, Sémin. Delange-Pisot-Poitou, 1967/68.
- [4] J. S. Balakrishnan, W. Craig, and K. Ono, *Variations of Lehmer's conjecture for Ramanujan's tau-function*, J. Number Theory **237** (2022), 90–104; arXiv:2005.10345. (States that Lehmer's conjecture remains open, and proves $\tau(n) \notin \{\pm 1, \pm 3, \pm 5, \pm 7, \pm 691\}$ for $n > 1$.)